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1 Introduction

2D Rotation Matrix and Vector Rotation

A rotation in two-dimensional space by an angle θ (measured counterclockwise) about the origin can be represented using the rotation matrix:

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Let the original vector be

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

When this vector is rotated by an angle θ , the new vector $\mathbf{v}'$ is obtained by multiplying the rotation matrix with the original vector:

$$\mathbf{v}' = R(\theta)\mathbf{v} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}.$$

Hence, the rotated vector $\mathbf{v}'$ has coordinates

$$(x', y') = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta).$$

Transforming Position Error from Global Frame to Robot Frame

Let the position error in the global (world) frame be represented as

$$\mathbf{e}_g = \begin{bmatrix} dx \\ dy \end{bmatrix},$$

and let the robot's orientation (heading) be given by the angle $\theta = \text{yaw}$, measured counterclockwise.

To express this error vector in the *robot's local coordinate frame*, we must rotate it by an angle $-\theta$ (i.e., the inverse of the robot's heading). This transformation is given by the rotation matrix:

$$R(-\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}.$$

Hence, the error vector in the robot's frame is:

$$\mathbf{e}_r = R(-\theta)\mathbf{e}_g = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix} = \begin{bmatrix} \cos \theta dx + \sin \theta dy \\ -\sin \theta dx + \cos \theta dy \end{bmatrix}.$$

The components of $\mathbf{e}_r$ are therefore:

$$\begin{cases} x' = \cos \theta dx + \sin \theta dy, & (\text{lateral error}) \\ y' = -\sin \theta dx + \cos \theta dy, & (\text{forward error}) \end{cases}$$

Thus, the **forward position error** (along the robot's local y -axis) is:

$$\boxed{\text{position_error} = \cos \theta dy - \sin \theta dx}$$

(*Note- You need to use x and y component of setpoint to calculate dx and dy)